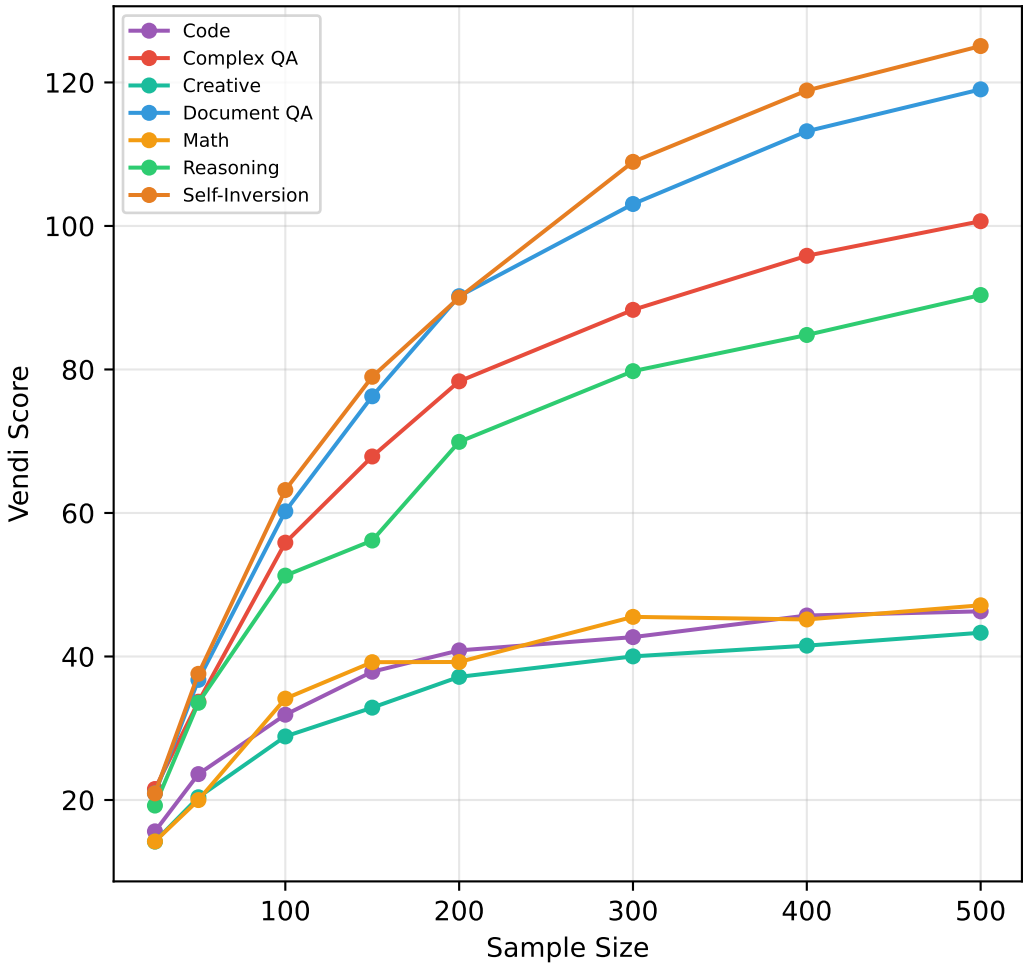
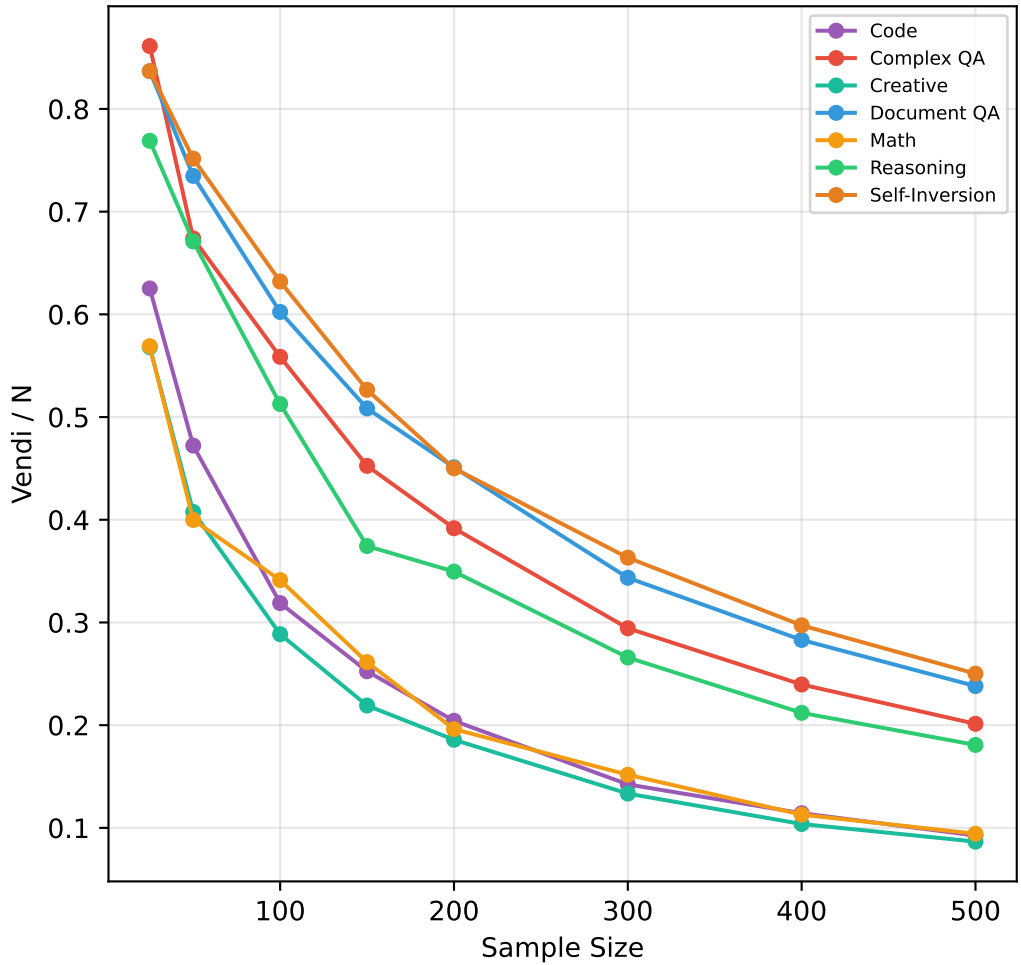


Fig 162: Per-Pipeline Contribution & Scaling Analysis

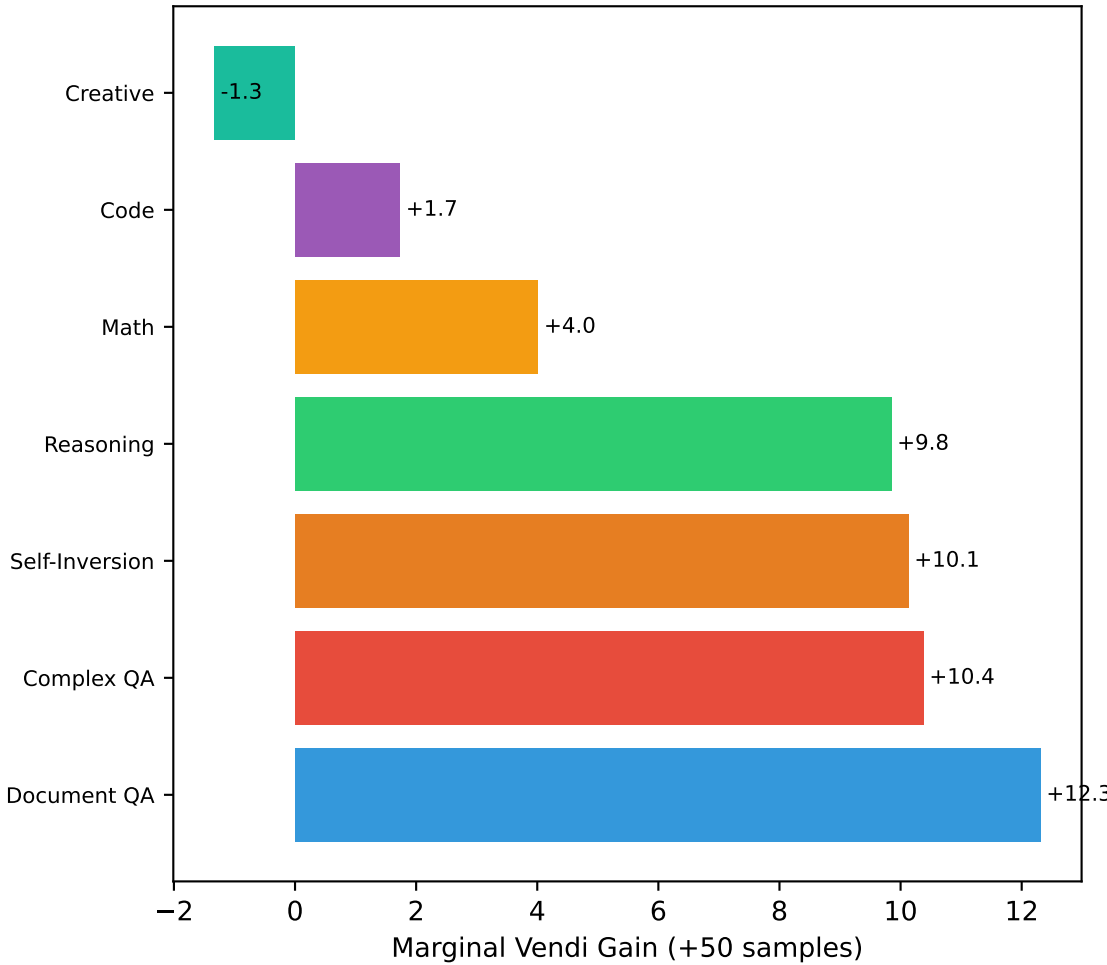
(a) Individual Scaling Curves



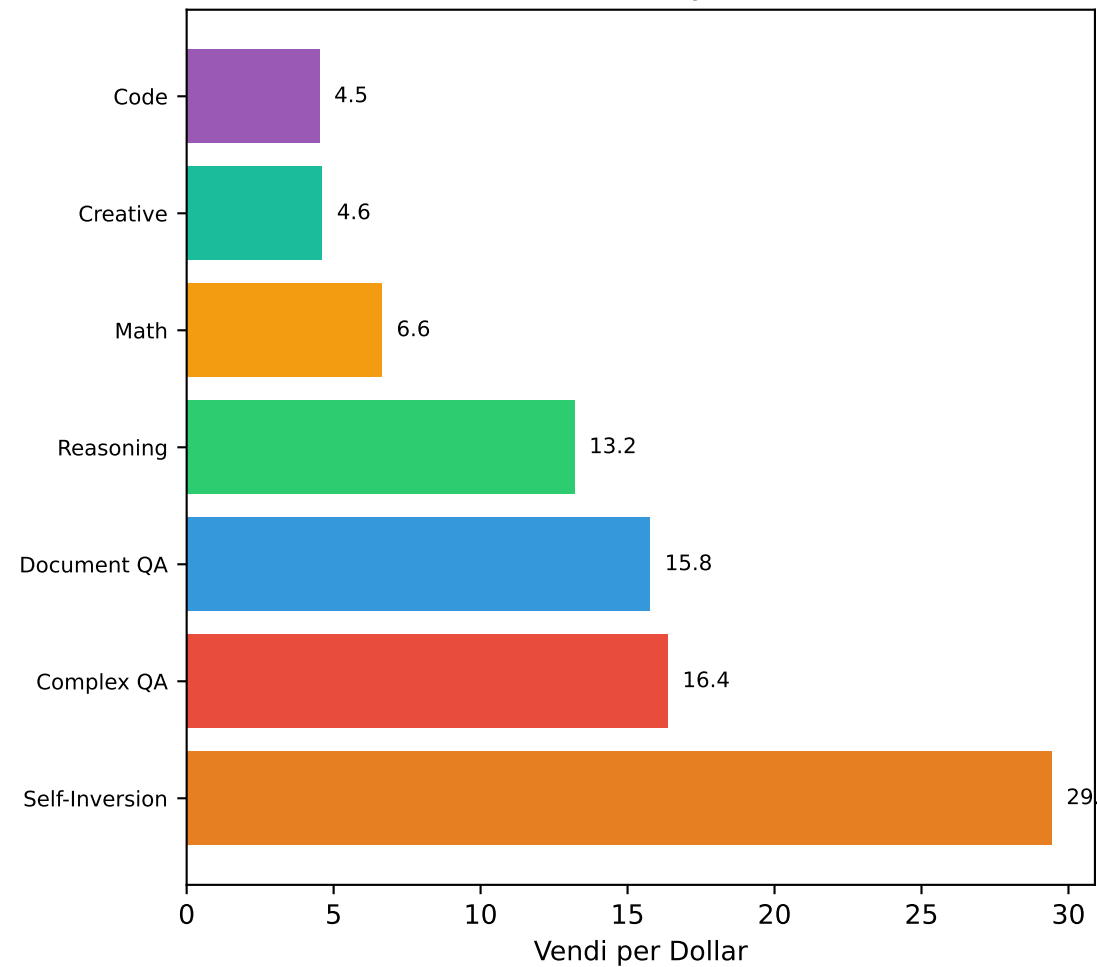
(b) Diversity Per Sample



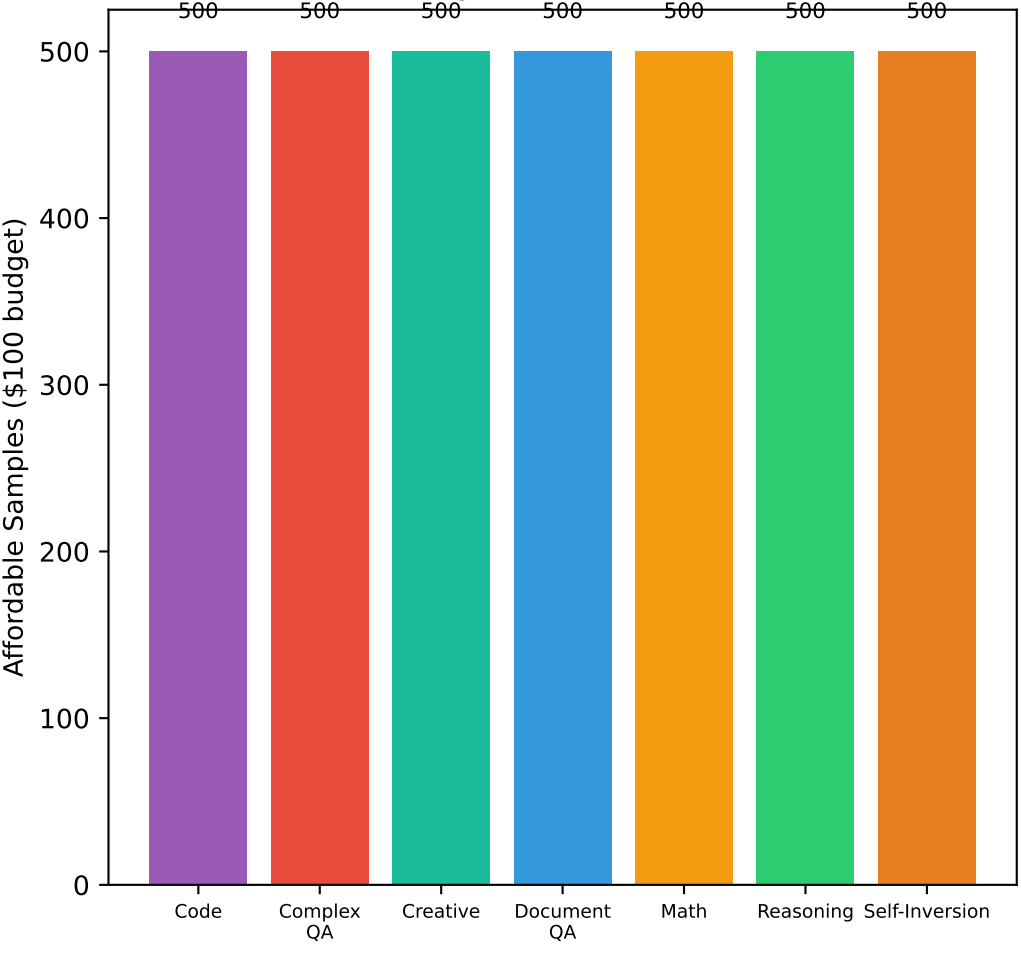
(c) Contribution to Random Base (Vendi=94.2)



(d) Cost-Efficiency (Vendi/\$)



(e) Budget Allocation (\$100)



(f) Budget Recommendations

Pipeline Contribution Analysis

Top contributors to global diversity:
Document QA: +12.3 Vendi
Complex QA: +10.4 Vendi
Self-Inversion: +10.1 Vendi

Base pool Vendi: 94.2

Most cost-efficient:
Self-Inversion: 29.4 Vendi/\$
Complex QA: 16.4 Vendi/\$
Document QA: 15.8 Vendi/\$

Recommendation:
1. Start with Self-Inversion (highest diversity, lowest cost)
2. Add Document QA (best synergy pair)
3. Add Reasoning + Complex QA (high contribution)
4. Code/Math for specialization
5. Creative only for niche tasks